Capture Query Statistics In Your Database

The queries in your database consume resources and it's helpful to focus your optimization efforts on the most costly queries. To do that you'll want to get a global view of all queries to see which ones are most costly. You'll use the pg_stat_statements[3] module which you configured in an earlier chapter, and will be referred to as PGSS.

PGSS takes each query and removes the parameters from it, creating a sort of query group or pattern. Each query then can be put into this group or pattern, and statistics can be collected at the group or pattern level. PGSS then makes the statistics available with a system catalog view.

These groups or patterns are called the *normalized* form of the query. Each normalized query gets a Query Identifier (queryid). The queryid uniquely identifies the original query text before the normalization process. The parameters are removed in the normalization process and replaced with placeholder characters.

Part of the purpose of the normalization where parameters are removed, is so that queries with the same structure can be grouped together. Statistics can then be computed across queries that are similar, sometimes referred to as a "group".

Query statistics are at the group level including the duration and the number of calls, across all instances of a query within the group that may have very different sets of specific parameters.

Enabling the PGSS module requires a database restart. PGSS is generally recommended to add for all databases. Although it adds a minimal amount of latency to collect statistics, the information it provides is worth it. PGSS is supported by cloud providers of PostgreSQL. If you don't already have it enabled in your production database, plan a time to make this parameter change and restart your database during a low activity period.

To configure the module, add pg_stat_statements to shared_preload_libraries in postgresql.conf. Once that's done and PostgreSQL has restarted, you'll enable the extension.

To enable the extension, run the following statement from psql. This only needs to be done once per database.

3. https://www.postgresql.org/docs/current/pgstatstatements.html